ASSIGNMENT-1

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19pu30

1. 1. Student and Social Security card

Each student can have at most one Social Security card and a social security card can be associated with atmost one student. So the relationship is

'One - one'

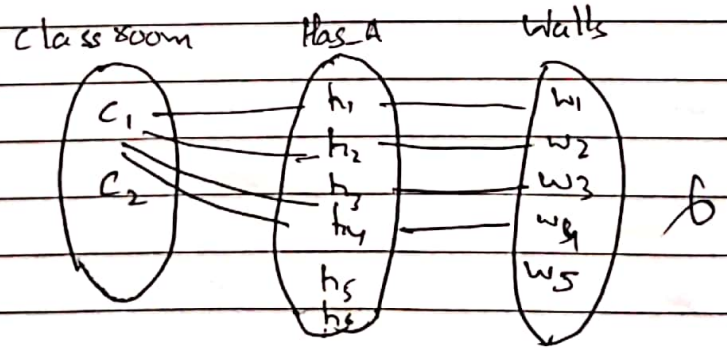
2. Student and Teachers.

A student can have many teachers and likewise a teachers can have many students. So relationship is

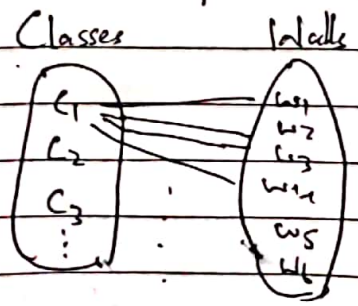
'many to many'

3. Classroom and wall.

Each classroom has multiple walls but a wall can be associated with only one classroom.



So the relationship is 'one to many'



Assumption: No two classroom are connected

4. Country & Current President.

Each country has only one current President and a person can hold the position of president for only one country. So the relationship is.

'One- One'

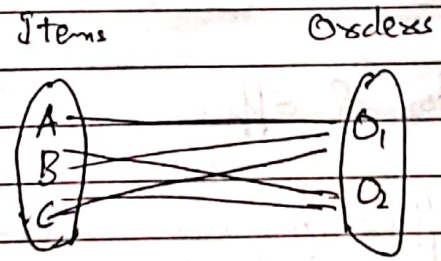
5. Course & Textbook.

Each course can have multiple text books but a text book is related to a single course. So the cardinality Ratio is

'One to many'

~~6.~~ 6. Item and order.

Consider 3 items A B C & 2 orders O_1 and O_2



From this we can say that a Item can be related to multiple orders and in the same way a order can be related to multiple Items so the Ratio is

'many to many'

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7. Student & Class.

If we assume class as subjects (DBMS, Mathematical foundation of Computing, Design and analysis of Algorithm etc) then each student can have multiple classes and each class will have multiple students so the ratio is

'many to many'

8. class & Instructor.

If we assume Instructor as the Teacher then and class to a section (Software System, DS, TCS etc) then each Instructor teaches different classes and each class is associated with different Instructors. The ratio is

'many to many'

9. Instructor & office.

Each office has one Instructor and each Instructor can instruct only one office so the relationship is

'one to one'.

10. E-bay Auction item & E bay bid.

For each one of E-bay Auction Item there are multiple Ebay bids so the ratio is

'one to many'.

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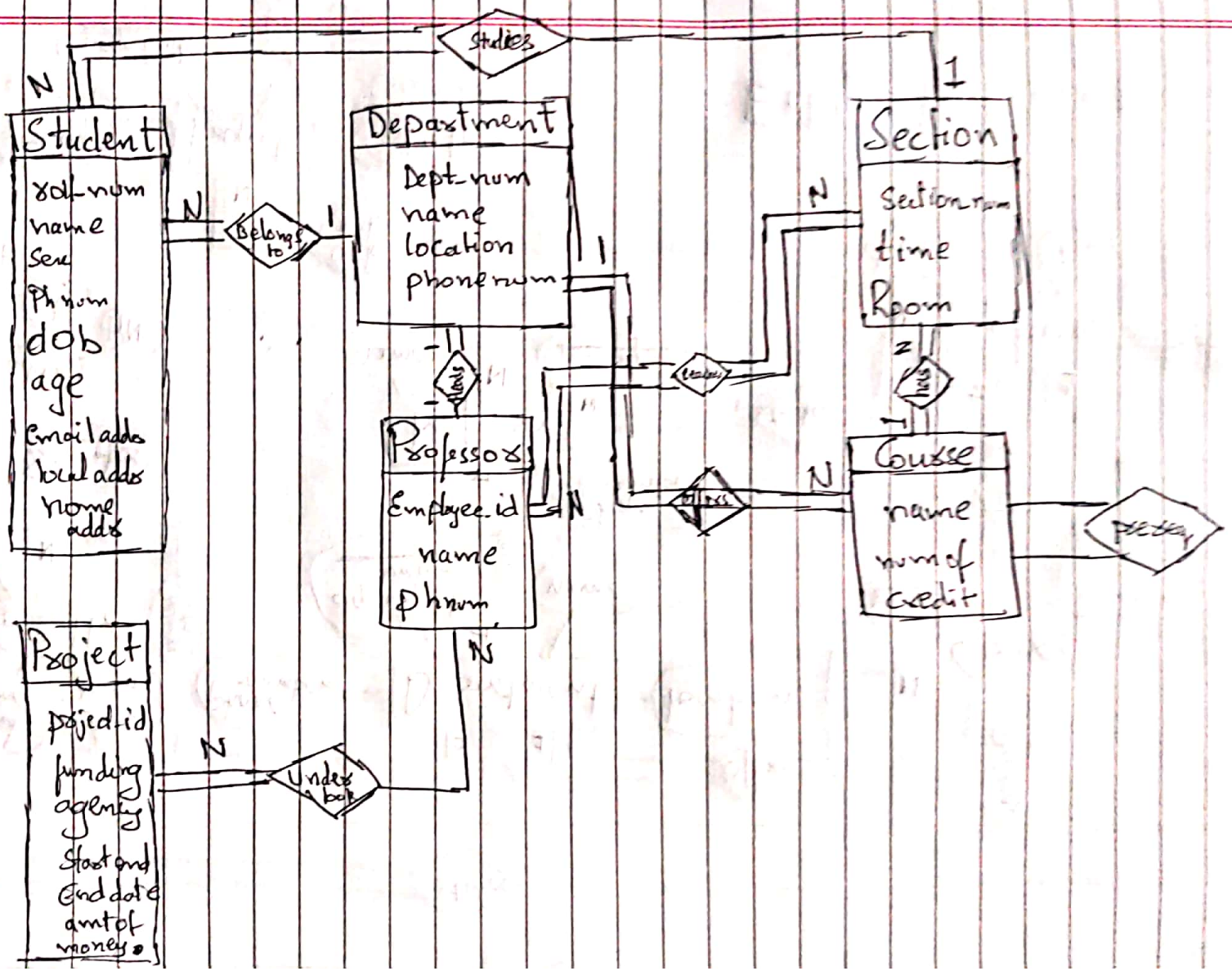
CONCEPTUAL SCHEMA



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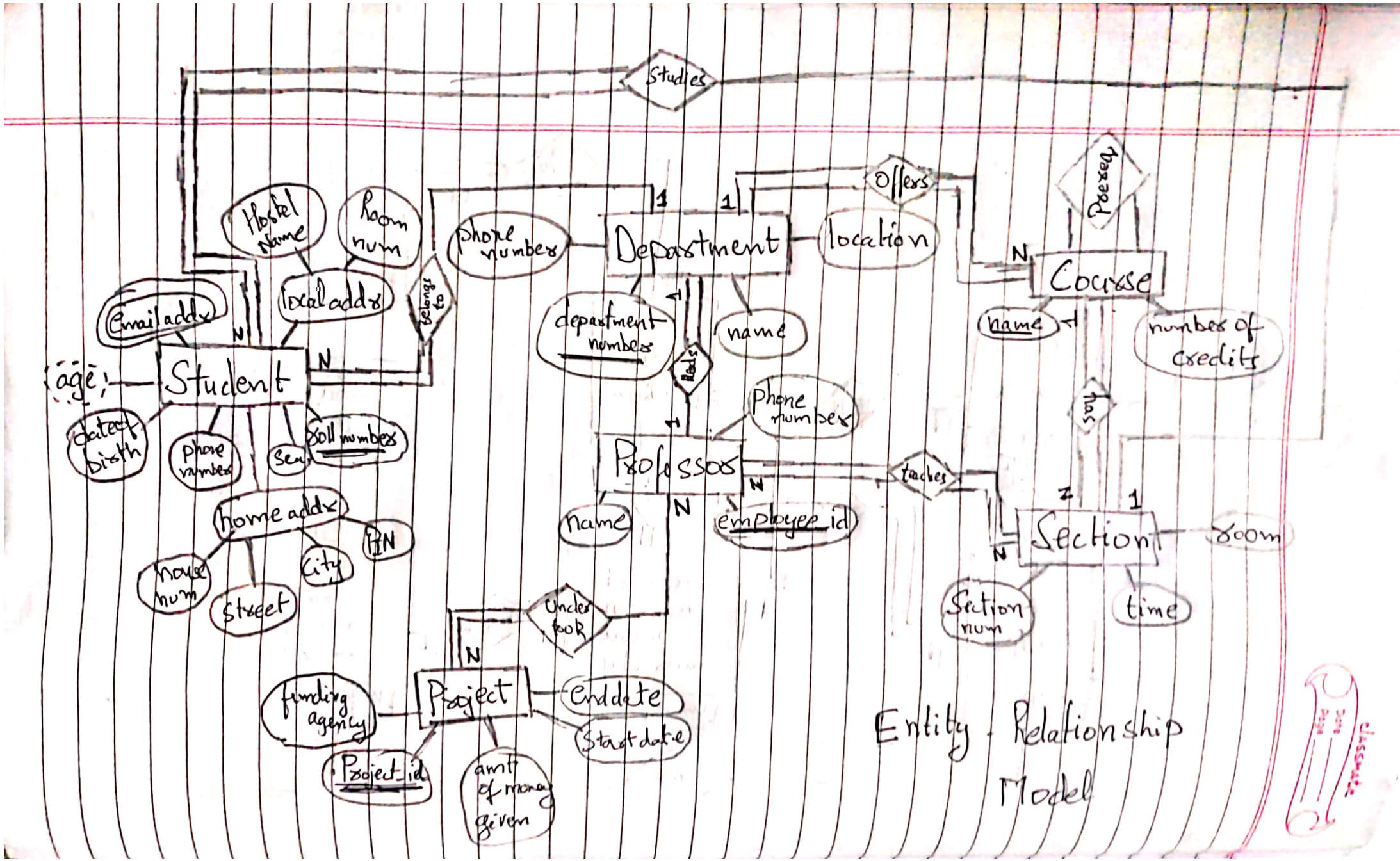

Entity - Relationship Model

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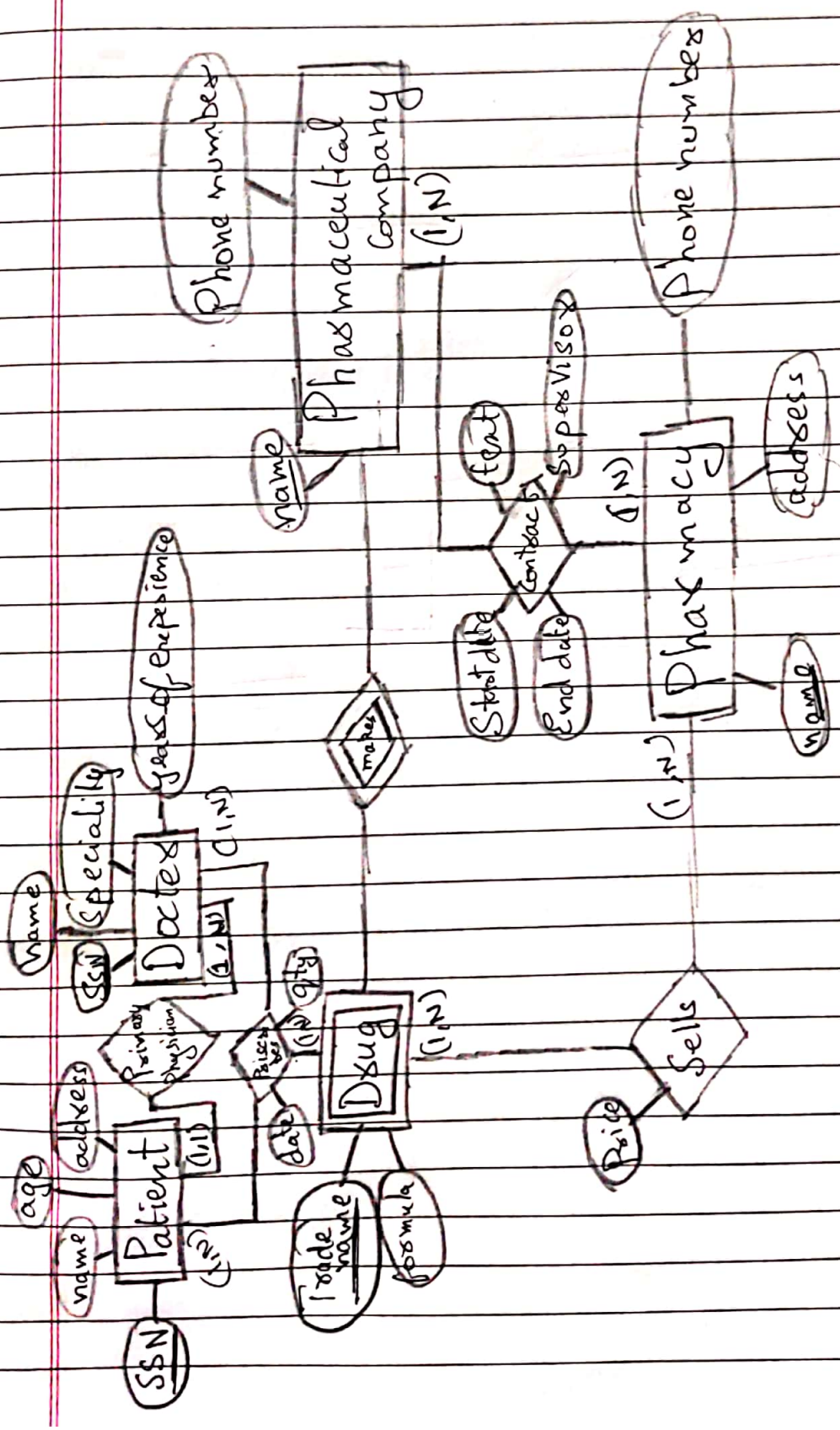
3. Entity Relationship Model
Prescription - R - X



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